1. Alert Management Practice

Objective

The objective of this task was to understand how Security Operations Center (SOC) analysts manage and prioritize security alerts. The task involved creating an alert classification system, prioritizing alerts, visualizing alerts in Wazuh, and documenting incident response procedures.

Tools Used

- **Wazuh** – Security Information and Event Management (SIEM) platform for monitoring alerts.
- **Google Sheets** – Used to classify and prioritize alerts.
- **TheHive** – Used for creating and managing incident tickets.
- **MITRE ATT&CK Framework** – Used for mapping attack techniques.

1) Alert Classification System

An alert classification system was created using **Google Sheets** to categorize security alerts based on their type, priority, and MITRE ATT&CK technique.

Example Classification Table

Alert ID	Alert Type	Priority	MITRE ATT&CK Technique
001	Phishing Email	High	T1566
002	Port Scan	Low	T1046
003	Malware Detection	Critical	T1204

This classification helps SOC analysts quickly identify the nature of an alert and respond accordingly.

2) Alert Prioritization

Alerts were prioritized based on their severity and potential impact. CVSS scores were used to determine the priority level.

Critical alerts require immediate investigation, while low-priority alerts are monitored for patterns.

Alert	CVSS Score	Priority
Log4Shell Exploit Detected	9.8	Critical
Brute Force Login Attempt	7.5	High
Port Scan	3.0	Low

	A	B	C	D	E
	Alert ID	Type	Priority	MITRE Technique	
	001	Phishing	high	T1566	
	002	SSH Brute Force	Medium	T1110	
	003	Log4Shell Exploit	Critical	T1190	
	004	Port Scan	Low	T1046	
	005	Phishing Email: Suspicious Link	High	T1566	

	A	B	C	D	E
1	Alert ID	Type	Priority	MITRE Technique	
2	001	Phishing	high	T1566	
3	002	SSH Brute Force	Medium	T1110	
4	003	Log4Shell Exploit	Critical	T1190	
5	004	Port Scan	Low	T1046	
6	005	Phishing Email: Suspicious Link	High	T1566	
7					
8					
9					
10					
11	Alert	CVSS Score	Priority		
12	Log4Shell Exploit	9.8	critical		
13	SSH Brute Force	6.5	medium		
14	Phishing Email	7.5	high		
15	Port Scan	3.1	low		
16					
17					

3) Wazuh Dashboard Visualization

A dashboard was created in **Wazuh** to visualize alert data. A **pie chart** was used to represent alert severity levels.

The visualization helps SOC analysts quickly identify:

- Number of alerts
- Severity levels
- Alert distribution

Example observation from the dashboard:

- Most alerts were **Level 3 alerts**
- A smaller portion consisted of higher-severity alerts.

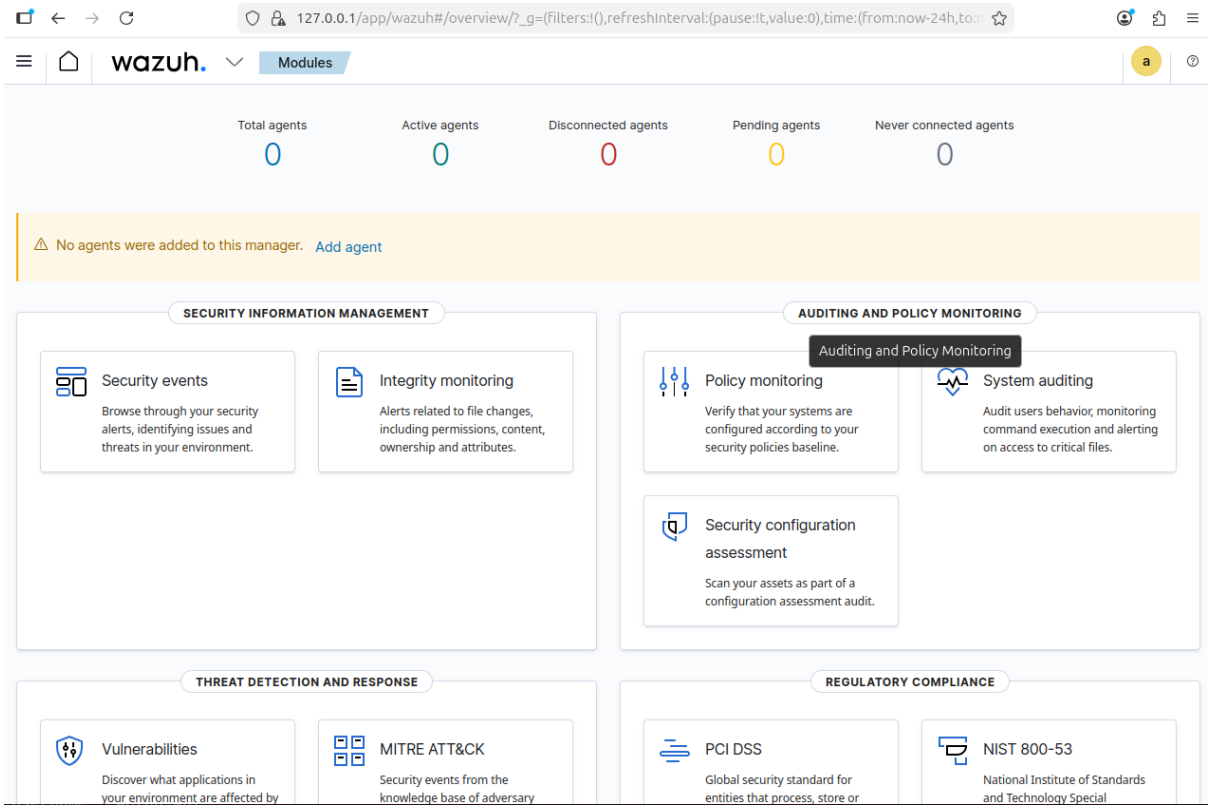

Fig: Dashboard of Wazuh

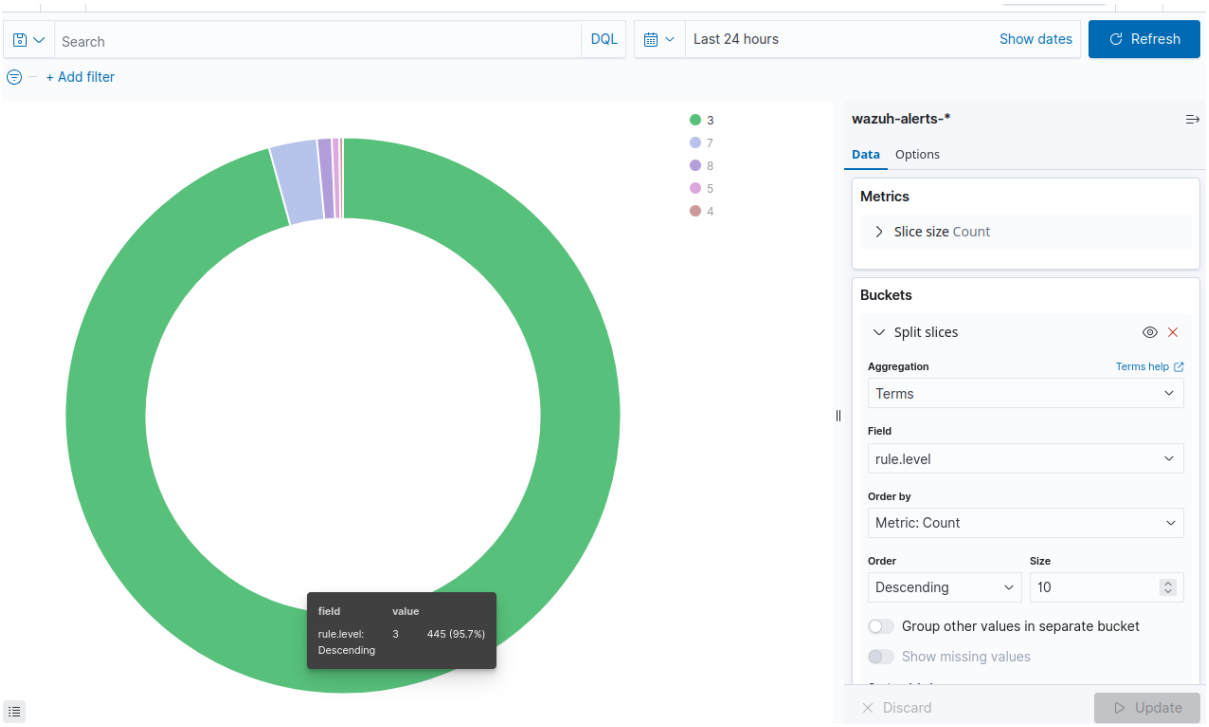

Fig: Wazuh Alert Severity Distribution by Rule Level

4) Incident Ticket Creation

A simulated incident ticket was created using **TheHive** to document the security incident.

Example Incident Ticket

Title:
Critical – Ransomware Detected on Server-X

Description:
Suspicious ransomware activity detected on the server. Indicators of compromise were identified during monitoring.

Indicators of Compromise (IOCs):

- File: crypto_locker.exe
- IP Address: 192.168.1.50

Priority: Critical

Assigned To: SOC Analyst

5) Escalation Procedure

A simulated escalation email was drafted to escalate a critical security incident to **Tier 2 SOC analysts**.

Example Escalation Email

Subject: Escalation – Critical Security Alert Detected

Dear Tier 2 SOC Team,

A critical security alert has been detected on Server-X indicating potential ransomware activity. Initial investigation identified the suspicious file crypto_locker.exe and communication with the IP address 192.168.1.50.

The alert has been classified as **Critical** due to its high CVSS score and potential impact on the system.

Further investigation and containment actions are required. Please review the incident and initiate the necessary response procedures.

Regards,
SOC Analyst

2. Response Documentation

Task 2 – Response Documentation

Objective

The objective of this task was to understand how security incidents are documented and handled in a Security Operations Center (SOC). This includes writing an incident response report, documenting investigation steps, creating a phishing investigation checklist, and summarizing lessons learned from the incident.

Incident Response

1. Executive Summary

During security monitoring, a suspicious email was detected that appeared to be a phishing attempt. The email contained a link that tried to redirect users to a fake login page. The SOC team investigated the email, checked the link reputation, and ensured that no users were affected. The malicious link was identified and blocked.

2. Timeline of Events

Timestamp	Action
2025-08-18 14:00	Suspicious email reported
2025-08-18 14:10	Email headers were checked
2025-08-18 14:20	Link reputation checked using VirusTotal
2025-08-18 14:30	Endpoint was isolated for safety
2025-08-18 14:40	Investigation completed

3. Impact Analysis

The phishing email attempted to trick users into clicking a malicious link. During the investigation, no users were found to have entered their credentials. Therefore, the impact was limited and no systems were compromised.

4. Remediation Steps

The following steps were taken to prevent further risk:

- The malicious link was blocked.
- Suspicious email was removed.
- Users were advised not to click unknown links.
- Email filtering rules were reviewed.

5. Lessons Learned

This incident showed the importance of monitoring suspicious emails and verifying links before opening them. Regular awareness training can help users identify phishing emails more easily and report them quickly.

Investigation Steps

During the investigation, the following actions were documented:

Timestamp	Action
2025-08-18 14:00	Suspicious email identified
2025-08-18 14:15	Link checked using VirusTotal
2025-08-18 14:30	Endpoint isolated
2025-08-18 14:45	Incident investigation completed

Phishing Investigation Checklist

The following checklist was used during the investigation:

- Check email headers
- Verify sender email address
- Check link reputation using VirusTotal
- Identify affected users
- Isolate suspicious endpoints
- Document the incident

The simulated phishing incident helped understand how security incidents are documented and handled in a SOC environment. The investigation process highlighted the importance of quick response, checking link reputation, and educating users about phishing attacks. Improving awareness and monitoring can help prevent similar incidents in the future.

3. Alert Triage Practice

We will simulate a **Brute-force SSH login alert**.

Example situation:

Many login attempts from the same IP trying different passwords.

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH Attempts	192.168.1.100	Medium	Under Investigation

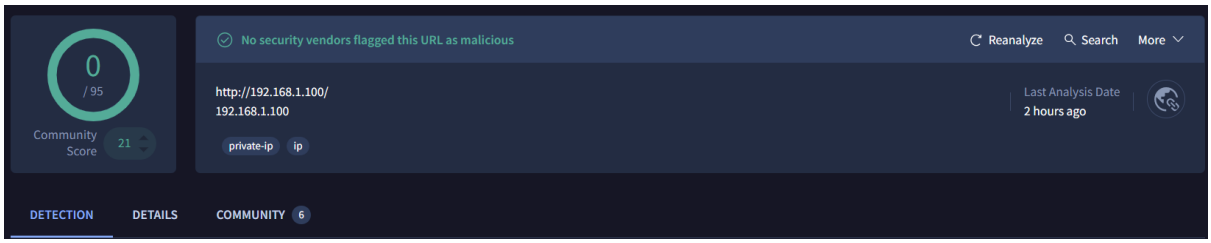

Figure: VirusTotal Threat Intelligence Analysis of Source IP Address

Threat Intelligence Validation

The IP address **192.168.1.100** was analysed using the threat intelligence platform **VirusTotal**. The scan results showed that **0 out of 95 security vendors flagged the IP address as malicious**. The IP is identified as a **private internal address**, which indicates that it belongs to a local network rather than the public internet. Based on this analysis, the alert was considered a **false positive**, but monitoring of the activity was recommended.

Therefore, the table is updated as below.

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH Attempts	192.168.1.100	Medium	False positive

The source IP address **192.168.1.100** was analysed using VirusTotal to validate whether it was associated with malicious activity. The scan results showed that none of the security vendors flagged the IP as malicious. Since it is a private internal IP address, the alert was classified as a **false positive**, but continued monitoring was recommended.

4. Evidence Preservation

SOC investigators must ensure:

- evidence is not modified
- evidence is traceable
- evidence has chain of custody

Without this, evidence is invalid in investigations.

Creating Evidence Storage Directory

A dedicated directory was created to store forensic evidence collected during the investigation.

Command used: mkdir evidence

cd evidence

netstat -tunap

Collecting Network Connection Evidence

Active network connections on the system were collected using the netstat command. This helps investigators identify active communications between the system and other hosts.

Command : netstat -tunap

Meanings:

-t	Displays TCP connections
-u	Displays UDP connections
-n	Shows numerical addresses
-a	Displays all active connections
-p	Shows associated process IDs

```
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:1514            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:1515            0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:5601          0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:55000           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:443             0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:39374         127.0.0.1:9200          ESTABLISHED -
tcp        0      0 127.0.0.1:41406         127.0.0.1:9200          ESTABLISHED -
tcp        0      0 127.0.0.1:41826         127.0.0.1:9200          ESTABLISHED -
tcp        0      0 127.0.0.1:41836         127.0.0.1:9200          ESTABLISHED -
tcp6       0      0 ::1:631                 :::*                    LISTEN      -
tcp6       0      0 127.0.0.1:9300          :::*                    LISTEN      -
tcp6       0      0 127.0.0.1:9200          :::*                    LISTEN      -
tcp6       0      0 127.0.0.1:9200          127.0.0.1:41836         ESTABLISHED -
tcp6       0      0 127.0.0.1:9200          127.0.0.1:41406         ESTABLISHED -
tcp6       0      0 127.0.0.1:9200          127.0.0.1:39374         ESTABLISHED -
tcp6       0      0 127.0.0.1:9200          127.0.0.1:41826         ESTABLISHED -
udp        0      0 127.0.0.54:53           0.0.0.0:*                           -
udp        0      0 127.0.0.53:53           0.0.0.0:*                           -
udp        0      0 192.168.170.128:68      192.168.170.254:67      ESTABLISHED -
udp        0      0 0.0.0.0:45228           0.0.0.0:*                           -
udp        0      0 0.0.0.0:5353            0.0.0.0:*                           -
udp6       0      0 :::48561                :::*                                -
udp6       0      0 :::5353                 :::*                                -
```

Figure: Active Network Connections Identified Using Netstat

The above figure shows the output of the netstat -tunap command used to identify active network connections on the system. This command displays currently established connections, listening ports, and associated network protocols. Collecting this information helps investigators understand ongoing communications and detect potential suspicious connections during an incident response investigation.

```
anthony-deepak@anthony-deepak-VMware20-1:~/evidence$
^C
anthony-deepak@anthony-deepak-VMware20-1:~/evidence$ sha256sum network_connectio
ns.txt
f8c346d996b4a6741f72b222579fbcfbe8a4f2ab3eda5024d6223cc2e7ce4210  network_connec
tions.txt
anthony-deepak@anthony-deepak-VMware20-1:~/evidence$
```

Figure: SHA256 Hash Generation for Evidence File

The above figure shows the SHA256 hash value generated for the evidence file network_connections.txt. Hashing is used in digital forensics to ensure evidence integrity. The generated hash value can later be used to verify that the evidence file has not been modified during storage or analysis.

Evidence Documentation

Item	Description	Collected By	Date	Hash Value
Network Connections	Active network connection data collected using netstat	SOC Analyst	05-03-2026	8c346d996b4a6741f72b222579fbcfbe8a4f2ab3eda5024d6223cc2e7ce4210

The collected network connection evidence was preserved by storing the output of the netstat command in a file named **network_connections.txt**. To ensure the integrity of the collected evidence, a **SHA256 hash value** was generated. The hash value acts as a digital fingerprint of the file and helps verify that the evidence has not been modified or tampered with during storage or analysis.

Maintaining evidence integrity is a critical step in **digital forensics and incident response**, as it ensures that the collected data remains reliable and can be used for further investigation or legal purposes.

5. Capstone Project: Full Alert-to-Response Cycle

Ubuntu → Metasploitable → Wazuh detects

In **VMware**, power on:

Metasploitable2

Find Target IP Address

Inside Metasploitable2 terminal run:

Ifconfig (command)

Role	Machine	IP
Attacker	Ubuntu	192.168.170.128
Victim	Metasploitable2	192.168.170.129

Scan the Victim Machine

Now using **Nmap**.

In Ubuntu:

Command: nmap -sV 192.168.170.129

```
anthony-deepak@anthony-deepak-VMware20-1:~$ nmap -sV 192.168.170.129
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-06 10:05 IST
Nmap scan report for 192.168.170.129
Host is up (0.0027s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.15 seconds
```

Figure: Nmap Service Version Scan Results

The Nmap tool was used to scan the target Metasploitable2 machine to identify open ports and running services. The scan revealed multiple open ports including FTP, SSH, HTTP, and MySQL. The FTP service was running **vsftpd 2.3.4**, which is a known vulnerable version that can be exploited.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.170.129:21 - The port used by the backdoor bind listener is already open
[+] 192.168.170.129:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.170.128:41207 -> 192.168.170.129:6200) at 2026-03-06 10:30:44 +0530
```

The Metasploit Framework was used to exploit the vulnerable VSFTPD 2.3.4 service running on the Metasploitable2 machine. After executing the exploit module exploit/unix/ftp/vsftpd_234_backdoor, a command shell session was successfully opened. The attacker gained root-level access to the target system, demonstrating successful exploitation of a public-facing vulnerability.

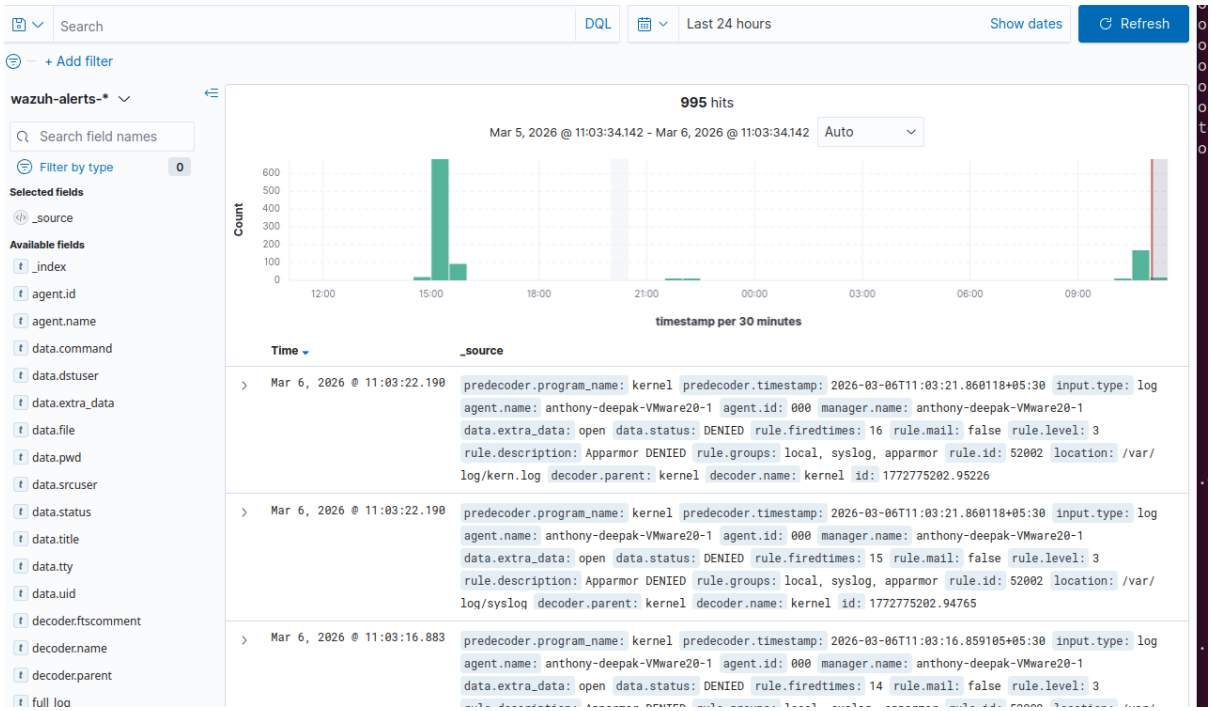

Figure: Wazuh Security Events Dashboard Showing System Security Logs

rule.description: AppArmor DENIED

location: /var/log/kern.log

rule.level: 3

The Wazuh dashboard was used to monitor security events generated from the monitored system. The dashboard displays system logs and alerts collected by the Wazuh manager, allowing analysts to investigate suspicious activities. In this case, multiple security logs such as AppArmor access denials and kernel events were observed, demonstrating how Wazuh collects and analyzes system-level security information for threat detection and monitoring.

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-06 10:30	192.168.170.128	VSFTPD Backdoor Exploit Attempt	T1190

The Wazuh SIEM platform was used to monitor security logs and identify suspicious activities occurring in the monitored environment during the simulated attack scenario.

Incident Report

Executive Summary

A simulated cyberattack was performed to understand how a Security Operations Center (SOC) detects and responds to security incidents. The goal of this exercise was to identify a vulnerable service, exploit it in a controlled environment, and observe how security monitoring tools can help detect and analyze suspicious activity.

Incident Description

During the simulation, a vulnerability scan was carried out on the Metasploitable2 virtual machine using the Nmap tool. The scan revealed several open services on the system. One of these services was an outdated FTP service called **vsftpd 2.3.4**, which is known to have a backdoor vulnerability. Using the Metasploit framework, this vulnerability was exploited successfully, which allowed remote command access to the target system with root-level privileges.

Detection and Analysis

The system activity was monitored using the **Wazuh SIEM platform**. Wazuh collected logs from the system and displayed them on the dashboard for analysis. These logs included kernel and AppArmor security events. By observing these logs, analysts can identify unusual or suspicious activities happening in the system.

Response and Mitigation

After identifying the simulated attack, the affected system was considered isolated to avoid further compromise. In a real-world scenario, the attacker's IP address would be blocked using firewall rules or other network security controls.

Conclusion

This exercise demonstrated how security teams can detect vulnerabilities, monitor system activities, and respond quickly to potential security threats. Continuous monitoring and proper incident response are important to protect systems from cyberattacks.

Manager Briefing

A simulated cybersecurity incident was carried out to test how well a system can detect and respond to possible attacks. During this exercise, a vulnerable service on a test machine was identified and intentionally exploited using security testing tools. The activity was monitored using the Wazuh security monitoring platform, which helps track system logs and detect unusual behaviour. This exercise showed how security teams can identify weaknesses, monitor suspicious activity, and respond quickly to security incidents. In a real situation, the affected system would be isolated and the attacker's access would be blocked. This simulation highlights the importance of continuous monitoring and quick response to protect systems from cyber threats.